

Analytical Coordinate Transformation for Manipulation when Using Robots to Serve People with Disabilities

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Abstract—The article presents an analytical approach for transforming the well-known Cartesian coordinates of details for manipulation into joint (photo-raster) coordinates. This transformation allows to minimize the time for programming robots and simplifies the requirements and cost of the robot control device. One of the examples, programmed according to the described approach, is realized when using robots to serve people with disabilities. The experience of the Institute of Robotics, Bulgarian Academy of Sciences in the creation of mobile service robots from the AnRI series (Anthropomorphic Robot with Intelligent) is also shown. They are designed to serve disadvantaged people and are built on the basis of a rolling cart with wheels. An application of Robot AnRI 1, which is equipped with a manipulator for serving disadvantaged people, is also shown.

Keyword: *service robots, mobile robots, robot control*

I. INTRODUCTION

Nowadays, more and more requirements are placed on the efficiency of the production of service robots. In recent decades, automation and modernization of production have served as a basis for the development of computer technology in this area and their very rapid development. The use of these technologies has a beneficial effect on the direct replacement of man by machine. At the present stage of development of technology, service robots designed to serve people with disabilities are gaining great importance in society's efforts to improve the comfort of life for these people. These robots are usually equipped with one or two manipulators to serve people with problems [3].

In robots such as manipulation ones three teaching methods are most commonly used: with the help of a manual teaching, using training console by puppet robot [1,2] or through a personal computer. The third way involves development and editing of programs by the computer. The results obtained from these programs are stored in the memory of the robot control device. In all cases, position of the robot is given in particular coordinates, which represent the number of pulses of the

incremental rotational sensors placed in different degrees of mobility. On the other hand, the coordinates of the points for selecting and placing the parts in a certain position are known in the Cartesian coordinate system and the training of robots is a difficult and time-consuming task. Therefore, it is necessary to generate an approach for the translation from a Cartesian coordinate system to a table of photo-raster pulses, which are joint, coordinates of the robot's axes.

II. TASK DESCRIPTION

The positioning of the robot links at two points A_1 and A_2 (Fig.1), the coordinates of which are known, or the distance between them is also known.

$$X_A = r_1 * \cos(Q_1) + r_2 * \cos(Q_1 + Q_2) - r_1 * \cos(Q_1 + q_1) + r_2 * \cos(Q_1 + q_1 + Q_2 + q_2) \quad (1)$$

$$Y_A = r_1 * \sin(Q_1) + r_2 * \sin(Q_1 + Q_2) - r_1 * \sin(Q_1 + q_1) - r_2 * \sin(Q_1 + q_1 + Q_2 + q_2) \quad (2)$$

$$\begin{aligned} q_1 &= (A_{12} - A_{11}) * 360^\circ / (I_1 + i_1) \\ q_2 &= (A_{22} - A_{21}) * 360^\circ / (I_1 + i_1) \\ XA &= XA_2 - XA_{21} \quad \text{and} \quad YA = YA_1 - YA_{21} \end{aligned} \quad (3)$$

where

r_1 and r_2 are the lengths of the robot links,

A_{11} , A_{12} are the number of pulses of the photo-raster rotation sensor of the first link when it is positioned at the target points, respectively, in point A_1 ;

and point $A_{21} = A_{22}$ – is the number of pulses of the second link photo-raster sensor of the second link when positioned at the target points, respectively, in point A_1 and point A_2 .

I_1 and I_2 – are the number of pulses per one turn photo-raster sensors.

i_1 and i_2 – are gear ratio correspondents to the first and the second links of the robot.

From equations (1), (2) and (3) we can form the following system:

$$r_1 * \cos(Q_1) + r_2 * \cos(Q_1 + Q_2) - r_1 * \cos(Q_1 + q_1) - r_2 * \cos(Q_1 + q_1 + Q_2 + q_2) - X_A = 0$$

$$r_1 * \sin(Q_1) + r_2 * \sin(Q_1 + Q_2) - r_1 * \sin(Q_1 + q_1) - r_2 * \sin(Q_1 + q_1 + Q_2 + q_2) - Y_A = 0 \quad (4)$$

with unknown angles Q_1 and Q_2 .

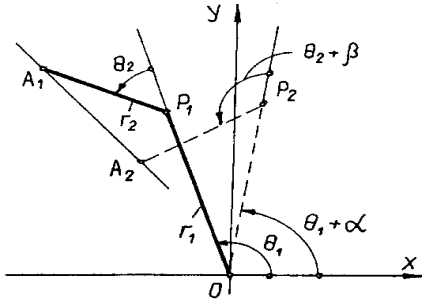


Fig. 1. Graphical presentation of the task.

For an approximate numerical solution of (4) we compose the matrix J:

$$\bar{J} = \delta F_i / \delta Q_j \quad i, j = 1, 2, 3, \dots \quad (5)$$

with elements

$$J_{11} = r_1 * ((\cos(Q_1 + q_1) - \cos(Q_1))) - r_2 * ((\cos(Q_1 + Q_2) + \cos(Q_1 + q_1 + Q_2 + q_2))),$$

$$J_{12} = r_2 * (\cos(Q_1 + q_1 + Q_2 + q_2) - (\cos(Q_1 + Q_2))),$$

$$J_{21} = r_1 * ((\sin(Q_1) - \sin(Q_1 + q_1))) * r_2 * ((\sin(Q_1 + Q_2) - (\sin(Q_1 + q_1 + Q_2 + q_2))),$$

$$J_{22} = r_2 * (\sin(Q_1 + Q_2) * (\sin(Q_1 + q_1 + Q_2 + q_2))).$$

If with $\vec{X} = (Q_1 + Q_2)^T$ we denote the unknown vector, then the Newton method [10] can be used for the numerical solution of systems of equations. In this case, in the initial approximation, each subsequent approximation is found by means of the solution of the equations [11]:

$$\vec{F} + (\vec{X}^{k+1} + \vec{X}^k) * \vec{J} = 0, \quad (6)$$

where

$$\vec{F} = (\vec{F}_1, \vec{F}_2)^T \quad (7)$$

To achieve convergence of the iterative process the following condition has to be must to fulfilled

$$\sum_{i=1}^2 \overrightarrow{F_{i+1}^2} - \sum_{i=1}^2 \overrightarrow{F_i^2} < 0, \quad (8)$$

To terminate the iterative process, the condition has to be fulfilled

$$\sum_{i=1}^2 \overrightarrow{F_{i+1}^2} - \sum_{i=1}^2 \overrightarrow{F_i^2} < \varepsilon, \quad (9)$$

where $\varepsilon > 0$ is a predetermined sufficiently small number depending on the desired accuracy of the calculations [8].

The above approach is applicable to determine the photo raster coordinates of a robot with a planar structure, with two regional degrees of mobility in a plane parallel to the working one. The approach is shown in Fig. 2.

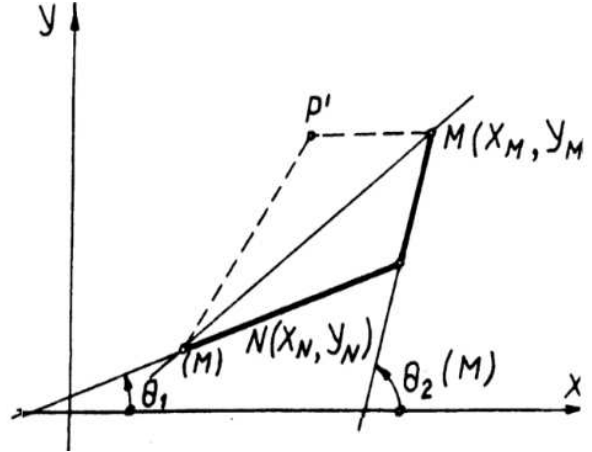


Fig. 2. Duality of possible solutions.

In order to determine the unknown number of photo raster pulses in the rotating pairs of the robot, it is first necessary to set the distance between the initial training points and the configuration of the robot. The second problem is their location of the point P, the location of the second rotary joint. According to Fig.2 the following system can be written:

$$\sqrt{(X_p - X_N) * (X_p - X_M) - (Y_p - Y_N) * (Y_p - Y_M)} = r_1 + r_2 \quad (10)$$

which has a solution if the following conditions are fulfilled:

$$\begin{aligned} (X_p - X_N) * (X_p - X_N) - (Y_p - Y_N) * (Y_p - Y_N) &= r_1^2 \\ (X_p - X_M) * (X_p - X_M) - (Y_p - Y_M) * (Y_p - Y_M) &= r_2^2 \end{aligned} \quad (11)$$

The resulting coordinate pairs reflect the two possible configurations of the robot links to reach a given point. The desired, position of the robot arms is determined by the sign of the vectors:

$$\overrightarrow{NP} \times \overrightarrow{PM} \quad (12)$$

Now is possible to determine the angles f1 and f2, of the robot links (r_1 and r_2) with respect to the coordinate system OXY. For this purpose, the orientations of the angles are determined $[0^\circ \text{ to } 2\pi]$, with given coordinates of the points $O(X_O, Y_O)$, $A(X_A, Y_A)$ and $B(X_B, Y_B)$. For assessment of orientation we introduce the conditional angles v_1 and v_2 :

$$\begin{aligned}
v_1 &= \\
&= \arcsin \left[\frac{\langle (Y_0 - Y_A) * (X_0 - X_A) - (Y_0 - Y_B) * (X_0 - X_B) \rangle}{r_1 * r_2} \right] \\
v_2 &= \\
&= \arccos \left[\frac{\langle (Y_0 - Y_A) * (Y_0 - B) - (X_0 - X_A) * (X_0 - X_B) \rangle}{r_1 * r_2} \right]
\end{aligned} \tag{13}$$

From equation (13) follows that

$$\begin{aligned}
f &= v_1 & \text{when} & & v_1 &= \pm v_2 \\
f &= 2n_i - v_1 & \text{when} & & v_1 &= n_i \pm v_2
\end{aligned}$$

After orientation and getting the right angles, it follows that those angles between the robot links and the OX axis reach the target point M

$$\begin{aligned}
f_1^M &= f_1 \\
f_2^M &= f_2 - f_1
\end{aligned}$$

As shown in Fig. 3, the proposed approach is suitable for installing integrated circuits in a straight line and for manipulating in matrices of elements in the service of people with problems such as arranging cups or cutlery.

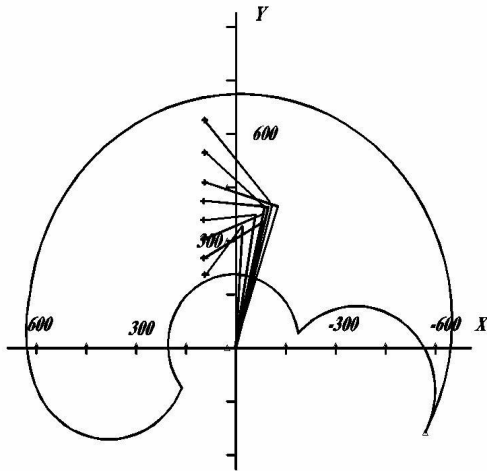


Fig. 3. Example of arranging integrated circuits or cups.

III. DESCRIPTION OF THE CREATED ROBOT ANRI

A. Requirements for Characteristics of AnRI Robots

Creating a service robot is a difficult and responsible task. The aim of this activity is to study the extent to which the developed robot will contribute to ensuring a dignified and independent life (with a focus on youth) for people with disabilities. The functional characteristics of the robot are

analyzed and designed in terms of the goals set for serving disadvantaged people. These are some specific characteristics of the robot manipulator, such as planning a safe trajectory in the context of the dynamically changing environment in which the serviced person lives [9].

To do this, robot use sensor information from built-in sensors such as "infrared" sensors and "ultrasonic" sensors, video sensor, and "End-Off" sensors to prevent collisions in the work environment. An opportunity has been created for building databases with the inclusion of libraries of ready-made programs for executing standard commands. In the experimental use of the robot, the results of its application are analyzed as the achievement of the set goals for serving disadvantaged people.

B. Reliability Requirements for AnRI Robots

In the operation of the robot for serving disadvantaged people, the requirements for reliability are increased in comparison with the standard requirements for the operation of industrial robots. As is well known, the human-machine interface is immediate and the robots work close to and in contact with the person being served. There is also an increased responsibility for the implementation of tasks for which there is no alternative solution and failure to do so can lead to unpredictable and sometimes tragic consequences. Otherwise, significant mental trauma may result. On the other hand, deviation from the normal operation of the robot can lead to specific injuries and damage to health in direct contact with the person served. That is why it is extremely important to create such methods of operation of the robot for serving disadvantaged people, which would guarantee the accurate and precise operation of these facilities. Of course, in addition to the described method of programming the robot manipulator, there are many other important elements of building its structure.

C. Description of the design of "AnRI-1" robot.

The robots of the "AnRI" (Anthropomorphic Robot with Intelligent) family (Fig. 4) are realized on the basis of constructions of mobile platforms with four wheels, two of which are driven directly and two are independent "free" wheels with the possibility of rotation around a vertical axis. The robot "AnRI -1" has a similar construction. It has the position of the wheels, which are mounted in the form of a horizontal "triangle". At the front in the two vertices of the triangle are the driven wheels, and at the third vertex at the back are the two paired "free" wheels with a distance between them of 150 mm. This makes the system stable and it is not possible to lose balance while moving. This wheeled layout scheme ensures that the mobile platform is easily manageable with increased mobility, which is especially important for the process of serving people with disabilities.

The hubs of the driven wheels have built-in electric motors powered by direct current from a rechargeable battery. "Worm" gearboxes are used for the driven wheels of the robot platform,

which do not allow reversing due to the built-in braking effect. In the upper part of the robot "AnRI -1" is mounted a manipulator type "hand" of anthropomorphic type with three regional and three local degrees of mobility.



Fig. 4: View of the Service Robot "AnRI-1"

The manipulator ends with a three-finger gripper with three separate independent drives [12,13]. On the basis of servo controllers with feedback from incremental sensors located in each degree of mobility, the drive of the robot manipulator is realized [14]. All regional connections are equipped with electromagnetic brakes [9].

D. Control System for AnRI Robots

The AnRI robot control system is of a hierarchical and distributed type. [6,7]. The connection between all modules is made via the serial interface RS 232. The processor module for robot control is a 32-bit microprocessor unit with the architecture of the Cortex-M processor family [12]. The Cortex-M4 family has proven to be very suitable for robot control applications. The processor module integrates a digital signal processing unit (DSP) with floating point support processing for a fast and energy efficient algorithm. Therefore, Cortex-M4 is used in digital control applications such as motor control, sensor information synthesis, power management and is suitable for the application of the proposed analytical coordinate transformation algorithm [5].

The implementation of the architecture of the robot control system "AnRI-1" is based on the processor module Cortex-M4. The created original I/O board for control of robots is used. With help of configuration of connecting different sensors and realizing robot drives are the control of a set of digital and analog inputs [2, 3 and 4]. The use of the modern architecture of the Cortex-M4 processor module simplifies the architecture of the AnRI-1 robot control system by reducing the number of external elements involved. This significantly improves the reliability of the entire system [2]. The use of Mixed Reality Server in the management system plays a special and very important role. This server allows the use of sensory information for both the state of the robot and the environment and operates safely based on a built-in database. The information is sent as an augmented reality stream to the supervisor's graphical user interface (GUI). There, part of the data processing takes place before it is sent to the user interface [1]. The ROS Bridge tool, in a ROS package, is used as part of the communication solution. Through ROS Bridge it is possible to access ROS through JSON formatted messages with one socket, and also through the so-called WEB socket. It uses TCP port (9090) for communication, which allows remote access and easy forwarding through network protections [1, 2]. WEB socket support is a new feature in the new versions of HTML 5 browsers, which is an advantage for the WEB-based version of the interface. In this way, WEB sockets provide a way to communicate with low latency, which is very important in such an application, where real-time response is crucial.

E. Experimental application and testing of Robot "AnRI-1"

The "AnRI-1" robot was tested and was available to people with disabilities. It was first tested to train users in the management process through basic operations, such as mobile platform movement and control of the individual levels of mobility of the manipulator. Thus, users received information about the capabilities of the robot. The next task was to perform purposeful movements. For example, at the user's command, bring water into a plastic bottle and fill a glass with a predetermined volume, the cup being delivered by the user (Fig. 6). The flow of running water from the bottle is a function of the inclination of the bottle, which is controlled by the robot. The volume of water poured into the glass is also determined. With a certain inclination of the bottle, an important parameter is the time for which the water flows into the glass, which is also controlled to prevent overflow.

The case in which the task was to deliver a book or magazine from the library to the serviced person is also interesting. At his command, after receiving instructions on which shelf and approximately where the requested book is located, the robot goes to the library. After finding it, the robot, with the help of its gripper, picks up the book from the library shelf and delivers it to the user. However, if the wrong book is delivered in a very small percentage of cases, the process is

repeated until the task is completed correctly. In the case of delivering a newspaper or magazine, the procedure is relatively easier, as these readings stand alone on the designated shelf and are relatively easy to find and deliver to the consumer.

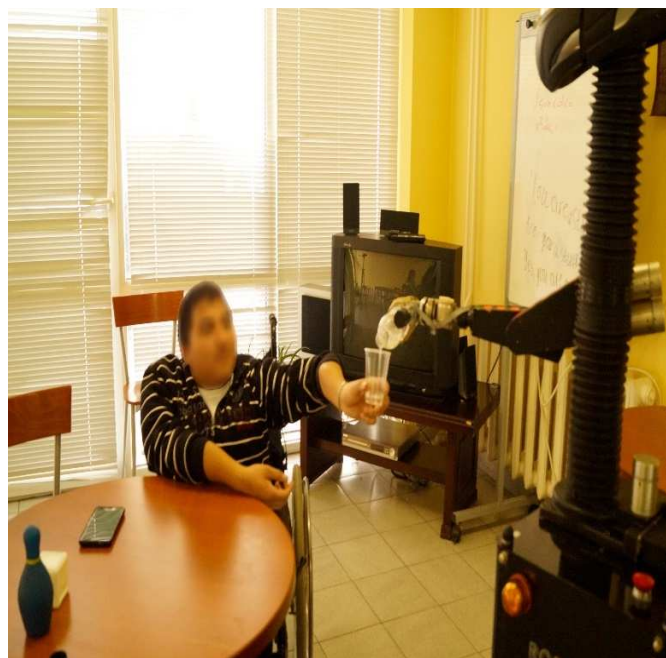


Fig. 5. Service Robot "AnRI-1" in action.

In the room where the experiments were conducted, it was used as a study room for reading and watching TV, so there was enough space for the robot to move. Of course, other algorithms for the robot's behavior can be created here, depending on the practical cases and needs, using the capabilities and tools available to the robot. The experimental research of the robot "AnRI-1" in the changing disorder is continuing, and at the same time the process of development of the robot's systems, mainly sensory-information and improvement of power (battery).

As a feedback from the users for the implemented project, an Attestation Card has been developed, in which social workers make suggestions in experiments with people from the pre-defined "Target Group" and so they can express their opinion on the use of the robot "AnRI-1" for serving disadvantaged people. Experiments with human service provided significant information about the peculiarities of the robot's behavior during work. Interruptions in connection with the exhaustion of the power supply possibilities should also be noted here. So monitor the condition of the battery and respond promptly. In case the capacity drops below a certain minimum, the robot should interrupt its work and go to the Docking Station installed in the room to restore the battery capacity. The robot can then continue to operate.

CONCLUSION

An analytical approach has been developed for transforming Cartesian coordinates of details for manipulation into articular (photo-raster) coordinates. This transformation allows to minimize the time of robot programming and simplifies the requirements and cost of the robot control device. The article also shows a model for controlling a service robot to optimize its functional characteristics, which ensures the reliability of service to the disabled person. The application of the proposed approach for programming the robot's manipulation system by transforming Cartesian coordinates into articulated coordinates allows the service robot to function reliably in the service of people, including people with disabilities. The control in the service process is carried out by recognizing voice commands, which is very important from the point of view of user convenience.

Based on the described data and the accumulated experience, it can be concluded that there is a growing tendency to create personal robots on wheels with one or two hands with three to five degrees of mobility or a tray for moving food, medicine or others. items to help mainly the elderly or disadvantaged. These service robots, in addition to the user knowledge base of the environment in which they work, need to be equipped with communication tools to communicate with the supervisor, as well as to assess critical situations related to the behavior and health of the serviced person. The proposed algorithm for controlling the robot manipulator to transform Cartesian coordinates of parts for manipulation into articulated (photo-raster) coordinates facilitates the user's activity. It can be summarized that the entry of service robots into people's lives will increase.

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